

Ultimate DSA Placement Dashboard - Software Requirements Specification (Version 1)

Objective

Build a single Excel workbook that acts as a complete placement preparation system for DSA. It should replace separate notebooks, trackers, planners, and revision sheets.

Target Users

Students preparing for Indian campus placements (especially 4-10 LPA roles) who want structured revision, analytics, and progress tracking.

Core Modules

1. Home Dashboard
2. Master Problem Database
3. Green/Yellow/Red Revision System
4. Smart Revision Planner
5. Pattern Tracker
6. Company Readiness Dashboard
7. Mock OA Tracker
8. CS Fundamentals Tracker
9. Interview Journal
10. Analytics & Charts

Master Problem Database

Each row should contain:

Problem Name, Topic, Pattern, Difficulty, ROI Score (1-10), Frequency (1-5 stars), Striver Link, LeetCode Link, GFG Link, YouTube Link, Attempt Date, Time Taken, Status (Green/Yellow/Red), Hint Used, Confidence (1-5), Revision Count, Next Revision Date, Notes.

Revision Rules

Green: Solved under 15 minutes without hints. Review after ~30 days.

Yellow: Solved with hint or took 15-30 minutes. Review after 7 days.

Red: Could not solve or required full solution. Review after 2 days.

Dashboard KPIs

Total Problems, Problems Solved, Green/Yellow/Red Counts, Revision Percentage, Weekly Progress, Monthly Progress, Average Solve Time, Placement Readiness Score.

Pattern Tracker

Track mastery for: Arrays, Hashing, Binary Search, Binary Search on Answer, Prefix Sum, Sliding Window, Two Pointer, Monotonic Stack, Linked List, Trees, BST, Heap, Greedy, DFS, BFS, Graphs, DP.

Company Readiness

Maintain readiness percentages for Accenture AASE, Cognizant GenC Next, Infosys DSE, Deloitte, Capgemini SSE, EY, Oracle and JPMC. Readiness should be driven by completion of high-ROI patterns and revision status.

Daily Planner

Auto-generate today's revision based on due dates. Mix topics instead of revising one topic at a time. Include 5-10 problems with balanced patterns.

Analytics

Provide charts for topic completion, pattern mastery, revision heatmap, average solving time, difficulty distribution, and overdue revisions.

Future Enhancements

Mock OA analytics, contest tracker, interview tracker, spaced repetition engine, achievement badges, and dynamic placement readiness model.

Design Principles

Minimal, professional UI; blue/white theme; clickable hyperlinks; filters; Excel tables; no duplicate problems; mobile-readable formatting.